
Uncertainty Quantification for Diffusion and Autoregressive LLMs: A Unified 200-Sample Benchmark on TriviaQA

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We present a compact but comprehensive experimental report for uncertainty quan-
2 tification (UQ) in large language models on open-domain question answering. We
3 benchmark seven UQ methods under one unified setup on TriviaQA (validation, rc
4 split): Mean Denoising Entropy, Final-Step Entropy, Trajectory Variance, Semantic
5 Entropy, Self-Consistency, P(True) elicitation, and Verbalized Confidence. We
6 evaluate three models with 200 examples per experiment: LLaDA-8B-Instruct
7 (diffusion LM), Meta-Llama-3-8B-Instruct (autoregressive LM), and Dream-v0-
8 Instruct-7B (custom autoregressive model with remote code). We report AUROC
9 for error detection and ECE for calibration. The report includes method definitions,
10 implementation details, compatibility fixes, complete tables, and reproducibility
11 information in one NeurIPS $\LaTeX$ file.

12 1 Introduction

13 Reliable confidence estimation is critical for deploying LLMs in retrieval, assistants, and decision
14 support. Prior UQ work mainly targets autoregressive (AR) models, while diffusion language models
15 (DLMs) expose richer intermediate trajectories that may carry additional uncertainty signals.

16 This report unifies UQ evaluation across DLM and AR models using one pipeline and one dataset
17 slice. Unlike a polished final paper, the goal here is to include all current project content in publishable
18 structure: problem statement, method taxonomy, protocol, results, observations, limitations, and
19 reproducibility artifacts.

20 **Scope.** We focus on short-answer TriviaQA and model-level uncertainty for answer correctness
21 (not token-level calibration on next-token prediction).

22 2 Problem Setup

23 Given a question x , model output $\hat{y}$, and binary correctness label $c \in \{0, 1\}$ from answer matching,
24 each UQ method outputs a scalar score u (uncertainty) or confidence $q \in [0, 1]$.

25 Evaluation.

- 26 • **AUROC:** ranking quality for detecting errors. For uncertainty scores, larger should indicate
27 wrong answers.
- 28 • **ECE:** expected calibration error using 10 bins on confidence in $[0, 1]$.

29 For uncertainty-only methods, confidence is mapped as $q = 1/(1 + u)$ in our implementation.

30 **3 Methods**

31 We evaluate seven methods from three experiment scripts.

32 **3.1 Trajectory UQ (single pass family)**

33 **Mean Denoising Entropy.** For DLM trajectory steps t and generated positions i , token entropy is

$$H_t^i = - \sum_v p(x_0^i = v \mid x_t) \log p(x_0^i = v \mid x_t). \quad (1)$$

34 Mean Denoising Entropy averages over masked positions and steps.

35 **Final-Step Entropy.** Entropy computed only at the last denoising step (or final AR step proxy).

36 **Trajectory Variance.** Fraction of positions whose argmax token changes across trajectory steps.

37 **3.2 Multi-sample semantic methods**

38 **Semantic Entropy.** Generate $K = 5$ answers; cluster by bidirectional NLI entailment; compute
39 cluster entropy:

$$H_{\text{sem}} = - \sum_c P(c) \log P(c). \quad (2)$$

40 **Self-Consistency.** Uncertainty is $1 - f_{\text{max}}$, where f_{max} is normalized frequency of the most
41 common normalized answer.

42 **3.3 Verbal confidence methods**

43 **P(True) elicitation.** Ask whether proposed answer is correct (Yes/No); use normalized probability
44 of Yes.

45 **Verbalized % Confidence.** Ask model to output “Confidence: X%” and parse $X/100$.

46 **4 Experimental Protocol**

47 **4.1 Dataset and matching**

48 TriviaQA validation (rc). Correctness uses normalized string matching with aliases and value fields.

49 **4.2 Models**

- 50 • **LLaDA-8B-Instruct** (GSAI-ML/LLaDA-8B-Instruct)
- 51 • **Meta-Llama-3-8B-Instruct** (meta-llama/Meta-Llama-3-8B-Instruct)
- 52 • **Dream-v0-Instruct-7B** (Dream-org/Dream-v0-Instruct-7B)

53 All reported runs below use **200 samples** per experiment.

54 **4.3 Key hyperparameters**

- 55 • Trajectory UQ: steps=64, gen_length=32
- 56 • Semantic experiment: $K = 5$, steps=32, gen_length=32, temperature=1.0
- 57 • Verbal experiment: steps=64, gen_length=32, vc_gen_length=64
- 58 • NLI model: cross-encoder/nli-deberta-v3-small

59 4.4 Implementation compatibility notes

60 To run all models in one codepath, we added AR/DLM branching and compatibility fixes:

- 61 • Support both DLM generation and AR autoregressive rollout in all scripts.
- 62 • Force boolean attention masks for Dream compatibility.
- 63 • Enable optional `trust_remote_code` for custom model repos.
- 64 • For Dream remote module, patched a `weights_only` incompatibility in downloaded module
- 65 code.
- 66 • Replaced reliance on `model.generate()` with manual AR loop where class lacks genera-
- 67 tion head support.

68 5 Main Results

69 5.1 Accuracy overview

Table 1: Task accuracy by experiment (all with 200 samples).

Model	Trajectory Exp.	Semantic Exp.	Verbal Exp.
LLaDA-8B-Instruct	0.280	0.280	0.255
Llama3-8B-Instruct	0.610	0.525	0.585
Dream-v0-Instruct-7B	0.250	0.150	0.295

70 5.2 UQ comparison table (AUROC / ECE)

71 5.3 Observations

72 **LLaDA.** In this 200-sample run, Mean Denoising Entropy is strongest by AUROC among LLaDA
73 methods and also best calibrated among non-verbal methods. Semantic Entropy drops versus
74 earlier 100-sample run, indicating sample-size sensitivity and potential variance from generation/NLI
75 clustering.

76 **Llama3.** Semantic and Self-Consistency are strongest discriminators. Verbalized confidence be-
77 comes more useful than in LLaDA (AUROC 0.6784), suggesting model-dependent verbal calibration
78 behavior.

79 **Dream.** Semantic and Self-Consistency are highest AUROC, but ECE remains high across most
80 methods. Verbalized-confidence parse rate is only 0.09, limiting practical use of that signal.

81 6 Analysis and Discussion

82 6.1 Cost-performance trade-off

- 83 • Single-pass family (trajectory metrics) is cheaper but may underperform semantic methods
- 84 on AR models.
- 85 • Multi-sample semantic family ($K \times T$ passes + NLI) gives strongest AUROC for Llama3
- 86 and Dream.
- 87 • P(True) often competitive but can be poorly calibrated depending on model.

88 6.2 Model-specific behavior

89 Different UQ families dominate different models. This suggests uncertainty should be treated as a
90 *model-algorithm pair*, not a one-size-fits-all metric.

Table 2: Unified UQ comparison. Each cell is AUROC / ECE (higher AUROC is better, lower ECE is better).

Method	LLaDA-8B-Instruct	Llama3-8B-Instruct	Dream-v0-Instruct-7B
Semantic Entropy	0.7667 / 0.2253	0.8222 / 0.1002	0.8875 / 0.3022
P(True) elicitation	0.7675 / 0.4237	0.7811 / 0.3400	0.8527 / 0.5364
Mean Denoising Entropy	0.7910 / 0.1057	0.7224 / 0.1622	0.7907 / 0.5258
Self-Consistency	0.7589 / 0.3592	0.8054 / 0.1867	0.8860 / 0.4643
Trajectory Variance	0.6290 / 0.4292	0.4936 / 0.1100	0.3815 / 0.4406
Verbalized % Confidence	0.5381 / 0.0617	0.6784 / 0.2318	0.5121 / 0.2153
Final-Step Entropy	0.5123 / 0.6857	0.5470 / 0.2859	0.6021 / 0.7453

6.3 Calibration caveat

Low ECE for verbalized confidence does not always imply high utility: if confidence range collapses or parse fails, discrimination can be weak despite favorable ECE.

7 Limitations

- Single dataset slice (TriviaQA rc validation) and small-to-moderate sample size (200).
- Exact-match style correctness can undercount semantically correct paraphrases.
- Semantic clustering quality depends on NLI threshold and model.
- Dream required remote-code compatibility patches; portability may vary with library versions.

8 Reproducibility

Core scripts.

- experiments/run_trajectory_uq.py
- experiments/run_semantic_entropy.py
- experiments/run_verbal_confidence.py
- run_experiment.sh

Result files used in this paper.

- LLaDA: results/triviaqa_traj_uq.json, results/triviaqa_sem_entropy_200.json, results/triviaqa_verbal_conf_200.json
- Llama3: results/meta-llama_Meta-Llama-3-8B-Instruct_triviaqa_traj_uq.json, ..._sem_entropy.json, ..._verbal_conf.json
- Dream: results/Dream-org_Dream-v0-Instruct-7B_triviaqa_traj_uq.json, ..._sem_entropy.json, ..._verbal_conf.json

9 Conclusion

This report consolidates all current UQ experiments into a unified NeurIPS-style artifact. Under consistent 200-sample evaluation, semantic-family methods dominate AUROC for Llama3 and Dream, while Mean Denoising Entropy remains particularly effective and efficient for LLaDA. Future work should extend to broader datasets, stronger semantic matching, and calibration transfer across model families.